Next Generation Hyper-Personalization Line (NGHPL)

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(Security / Sensitivity) Classification

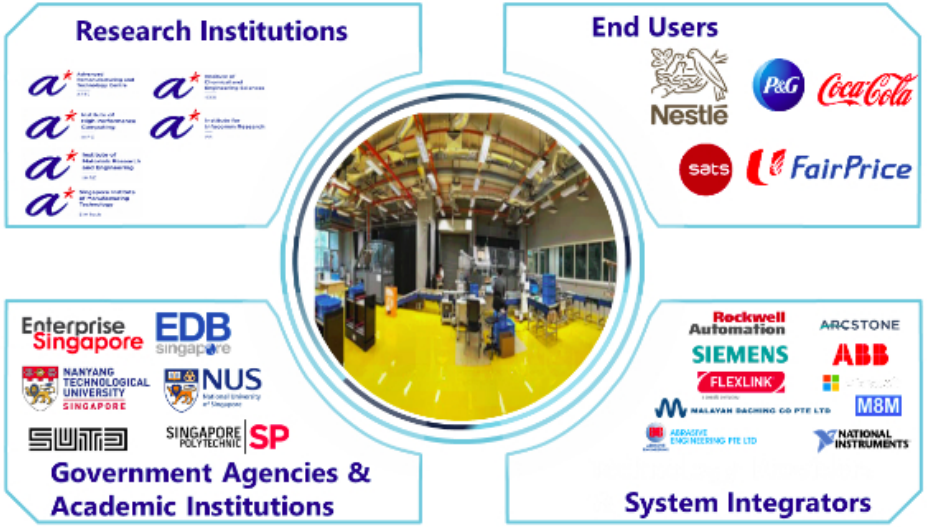
What is Hyper-Personalisation?

Hyper-Personalisation is a cross-sector consumer trend demanded by discerning tech-savvy customers and enabled by eCommerce and advances in technology

Examples of different levels of Personalisation	Adidas Shoes	
Mass Production	Adidas general purpose trainers	
Mass Customisation	Adidas walking trainers for female	
Personalisation	Adidas walking trainers for female with personalised colour, leather, heel toe	
Hyper-Personalisation "Batch size of one " "MOQ1"	Hyperpersonalised – Adidas shoes via speed factory, design your own shoes	

The consumer demand for Hyper-Personalisation is cross-sector, the Focus for the Next-Generation Hyper-personalisation Line is the Fast-Moving Consumer Goods Sector

3-Year NGHPL Testbed $ 20 million Open Platform



Outcome: Provide Hands-on Testing, Impart Knowhow and build a Digitally Savvy Workforce

Commenced in July 2019, 18 months complete

(Security / Sensitivity) Classification

Source/Reference:

Agency for Science, Technology and Research
SINGAPORE

Delivering a whole new customer experience

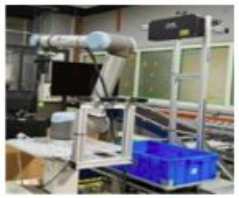
The Next Generation Hyper-Personalisation Line (NGHPL) is a smart, scalable & integrated modular test bed for hyper-personalised product manufacturing and order customisation. We develop solutions with industry to improve productivity and reduce time & cost to implementation.

Advanced Remanufacturing and Technology Centre
Institute of High Performance Computing
Institute for Infocomm Research
Singapore Institute of Manufacturing Technology
NUS National University of Singapore

Journey of Next Generation Hyper Personalization Line (NGHPL)

Repurposing to Include Biomedical Manufacturing Applications

3D Vision Robotic Bin Picking



Standalone

Multiple Size Carton Forming



Batch 1 Production Smart Mixing & Filling Line



Product Replenishment Line

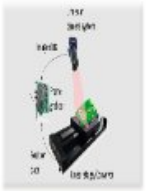


Order Fulfilment Line



Automation + End to End Connectivity

Smart Quality



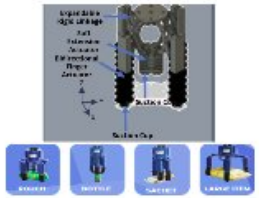
Smart Cleaning



Robotic Labelling and Bundling



Flexible Gripper Design



Robotic Mixed Palletizing



AI & ML

Gen AI Assisted Smart Maintenance



AMAv-X Line





MSP, CRP & Showcases

- 1 Start-up Company
- 21 Technology Licences
- 35 MSPs completed

- > 13 R&D Jobs created
- > 9 companies signed on as ARTC members

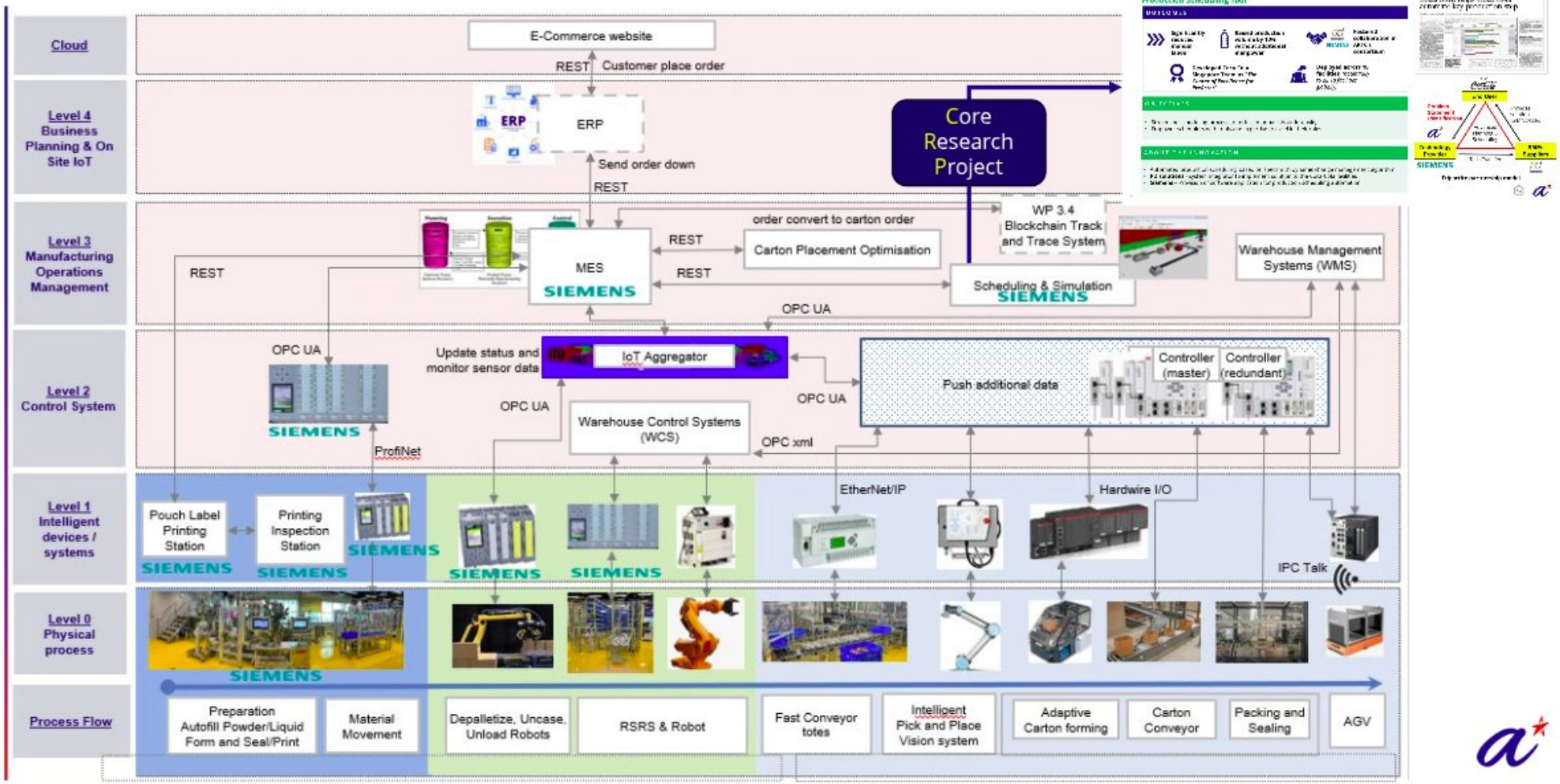
MSP, CRP & Gap Projects

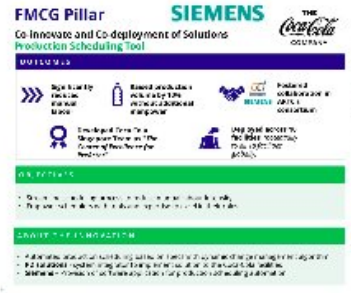
Pilot Trials MSP, CRP & Gap Projects

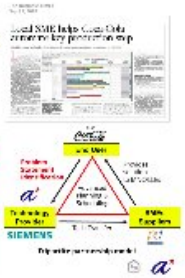
2018 Pre-Development (Seed Funding) 2019 Development and Commissioning of FMCG Line (IAF-PP NGHPL) 2022 Upgrading 2025 Revamp

Next Generation Hyper Personalization Line (NGHPL) System Architecture

Siemens Controllers and Software for the line







Collaboration between ARTC – Siemens – Coca Cola

Advanced Remanufacturing and Technology Centre
ARTC

THE Coca-Cola COMPANY

SIEMENS

Local SME helps Coca-Cola automate key production step

Scheduling system developed by PD Solutions and A*Star unit cuts time taken to mere minutes. BY LYNETTE TAN



Despite various challenges, the team from PD Solutions and A*Star unit ARTC was able to complete and implement the new system at the plant in less than six months.

PHOTO: COCA-COLA SINGAPORE

Opening new doors

Automation helps firms ease labour crunch, shorten processes



- Issues: Need to improve resource allocation, maximize utilization and prevents bottleneck in production
- Results:
  - Reduce scheduling time from 3-5 days manually to 10-15 seconds by using the software
  - Enhance the stability of the schedule due to less human intervention
  - Provide efficient management visualization capability
  - Strengthen the communication between management strategy and production activity
  - Allow efficiency in managing special order status, e.g. unscheduled, rework, missing
  - SG Factory and scale up to 7 other Coca Cola manufacturing sites across the globe over the next 18 months

(Security / Sensitivity) Classification

Source/Reference:

NGHPL Success Stories: Factory Translation & R&D Programmes



Mirxes Programme
a pioneering life science start-up driving automation, sustainability, and resilient supply chains.



WEF Lighthouse Contribution
RO-Plus, a robotic start-up developed soft agile gripping technologies supported the TCCC WEF Lighthouse use case.

Projects Delivery
Over 40 projects delivered with FMCG leading companies and on-going projects with Biomedical companies.



Supply Chain 4.0 Programme
Uncovered critical gaps in supply chain resilience, leading to the development of the Supply Chain 4.0 Programme.



Delivery Under Pandemic
On-Time Project Delivery during COVID for Automated Stronghold Lab Projects & Digital Solutions.



Modular Autonomous Lab (MAL)
Cutting-edge technology successfully deployed and utilized to power operations for industry lab automation.

COVID UTM END to END TRACKING & TRACEABILITY

SIEMENS
Ingenuity for life

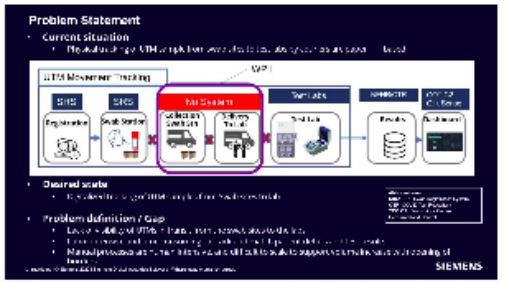
Background:
During early 2020, Singapore's Ministry of Health needed to scale PCR testing capacity from 1,000 to 10,000 UTMs/day
DISW proactively developed a working Mendix prototype within **2 days**, demonstrating Siemens' rapid solution capability.
In June 2020, the team pitched to multiple government agencies (AStar ARTC, SIMTech, MOH, HPB)

Challenges:
Capacity scaling: Tenfold increase in testing volume with **potential 4x** surge during peak periods
Critical errors: Public reports of swab samples mistakenly disposed before testing
Manual processes: Paper-based tracking prone to human error at high volumes
No visibility: Lack of real-time end-to-end sample traceability

Outcomes:
Phase 1 (Q2 2021): First app deployed at **2 sites** (Shu Qun Secondary School, Stronghold Diagnostics Lab)
Phase 2 (Q4 2021): Expanded to second app for COVID patient conveyance and hospital allocation

Solution:
Web App (Swab Sites): Upload sample details by boxes from Registration System
Mobile App (Couriers): Pickup/delivery tracking with turn-by-turn GPS monitoring
Mobile App (Labs): Quantity verification and mismatch reporting
Reports & Analytics: End-to-end traceability dashboards for authorities and labs

Impact:
Eliminated manual errors, enabled real-time tracking, and provided complete chain of custody from collection to testing





Unrestricted

Courier Pickup using mobile device

(i) Courier driver login to Mobile device

(ii) Driver selected Shujun1 site & SDL01 Lab → Shipment# auto-generated with timestamp

Driver update Swab site POC staff & contact

(iii) Driver scan box Qrcode to be collected

(iv) After scanning 4 boxes, Swab site sign-off the shipment

(v) Driver press Navigate to Lab











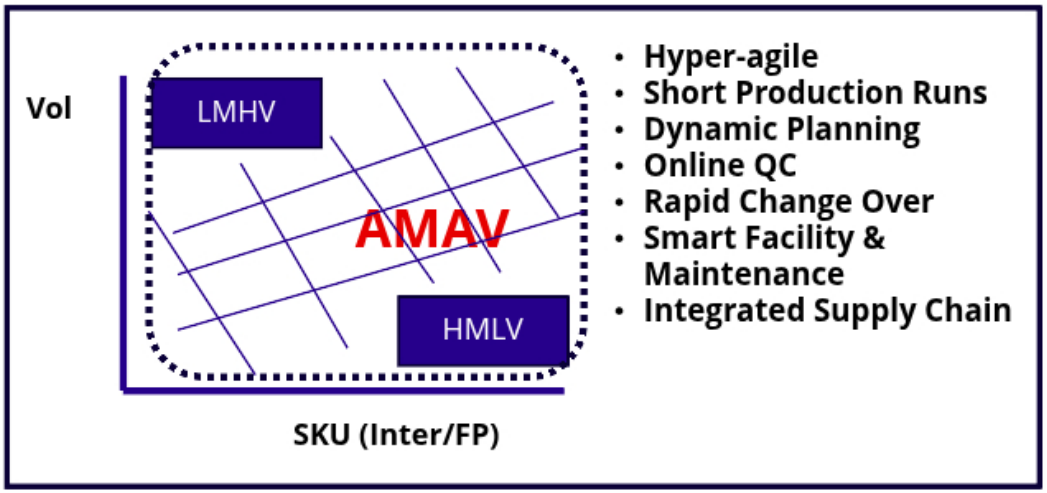
Any Mix Any Volume-X (AMAv-X) Line

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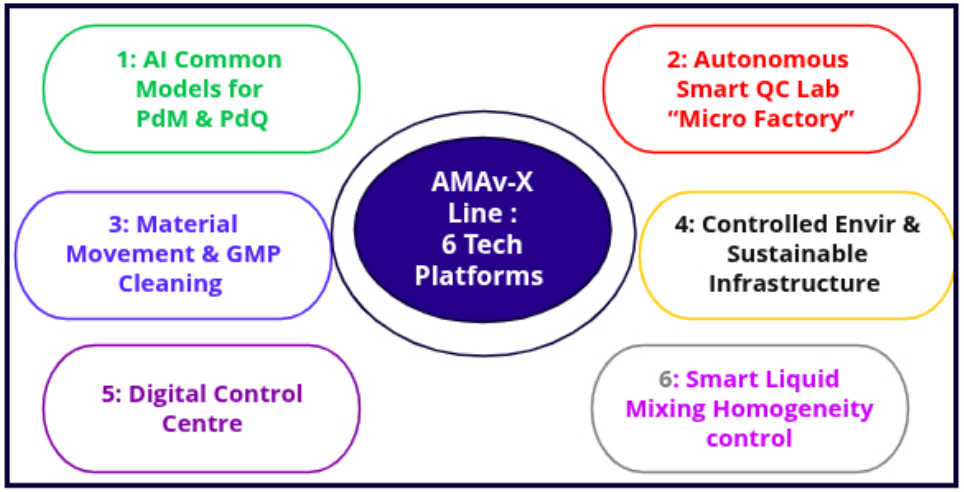


(Security / Sensitivity) Classification

Any Mix Any Volume-X (AMAv-X) Line for SSAM Capability Building



- Hyper-agile
- Short Production Runs
- Dynamic Planning
- Online QC
- Rapid Change Over
- Smart Facility & Maintenance
- Integrated Supply Chain





1. For Any random mix of volume, Small batch size production, Limited historical data and Minimum number of IOT sensors



2. For Manual lab operation, with large repetitive sampling with strict quality compliance, requiring human disposition of results



3. For Manual High Frequency Cleaning and Validation, Experienced- based operation & Manual cleanroom control



4. For cleanroom operations are controlled based on the objective to maintain air changes / hour which can result in overcooling and excess energy consumption



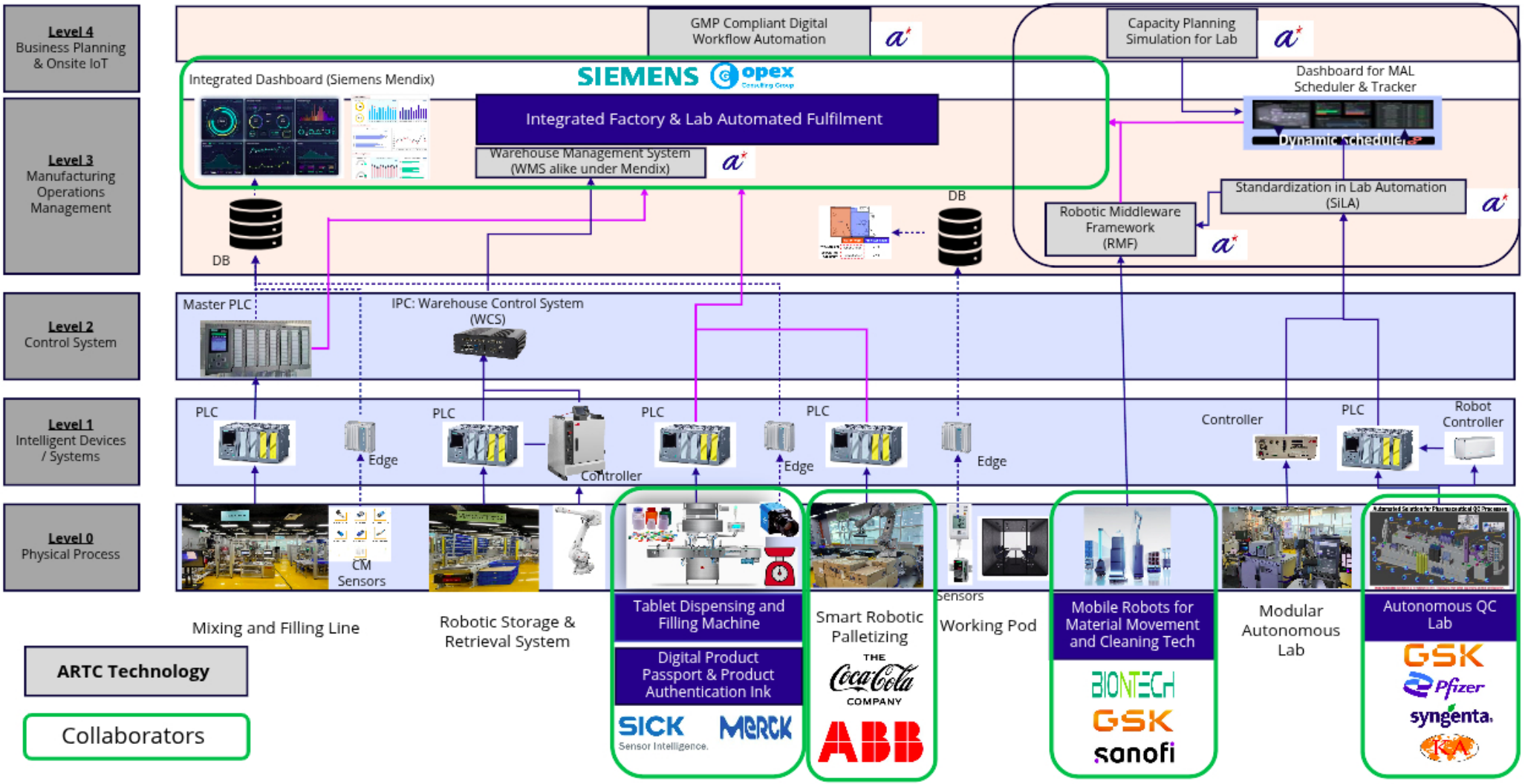
5. For manual work orders placement at machine level and no real-time work order data collection / tracking



6. For liquid mixing process that need extensive study and validation to determine mixing time required for each liquid formulation in both production and lab

X = multi-industry, manual or automated, continuous or batch processes

System Architecture for Any Mix Any Volume-X (AMAv-X)



Any Mix Any Volume-X (AMAv-X) Line : Phases 1 and 2

Goal:
1. Establish an autonomous, modular QC lab to enable "Lab of the Future"
2. Deploy next-gen tech to create a flexible, connected "Factory of the Future"
3. Use AI, robotics & IoT to drive smart, sustainable human–machine collaboration
4. Showcase S&A's end-to-end capabilities via real use cases in Biomedical & FMCG Pillars

Business Value:
1. Speed, precision, & risk mgt QC operation & decision
2. E2E Productivity Advanced Mfg Technologies
3. Regulatory Compliant Automation & Digital Solutions
4. Towards Net Zero Sustainability Mgt Capability

Zone #1



Batch-X Manufacturing (Pouches)
1) Personalized pouch production
2) Pouch Leakage Inspection
3) AI empowered Predictive Maintenance with Limited Data

Zone #2

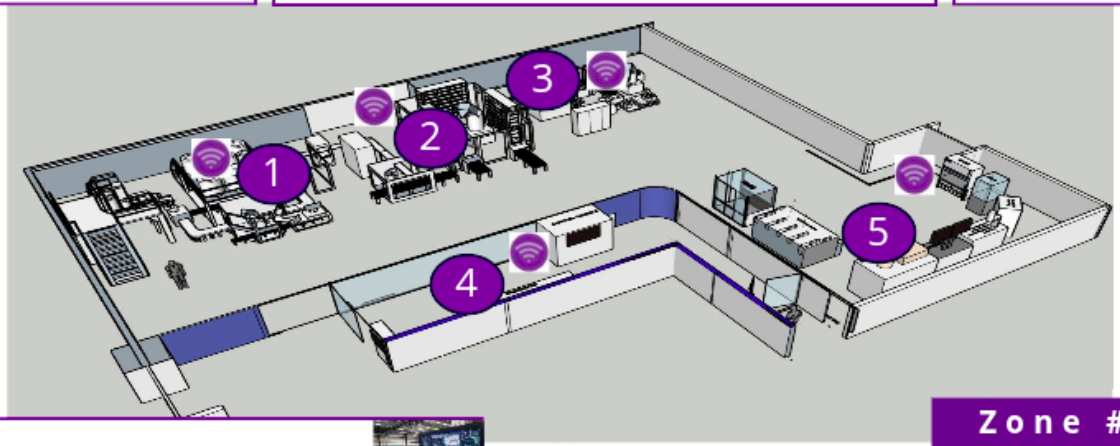


Material Movement and Cleaning Technology
1) Robotic Storage and Retrieval System
2) RMF Multi-robot fleet management for material movement and cleaning
3) GMP compliant robotic cleaning technology (Floor & Wall)

Zone #3



Continuous Processing Line (Tablet Dispensing)
1) Robotic Mixed Palletizing
2) Continuous tablet dispensing and filling line
3) AI empowered detecting and predicting product quality
4) Smart Liquid Mixing Homogeneity control



Zone #4



Digital Control Centre
1) GMP compliant Digital Workflow Automation and AI Analytic
2) Low-Code No-Code End-to-End Connectivity and Integrated Dashboard: Production Tracking Dashboard, AI of Predictive Maintenance & Predictive Quality Dashboard, ECMAS Dashboard, MAL Dashboard

Zone #5



Automated Green Lab
1) Automated High-speed Pill Dicer
2) RAISE Machine, AVID, Automated Liquid Handling
3) Modular Autonomous Lab (MAL) + Dynamic Scheduler and Tracker
4) Autonomous QC Lab + AI GC Data Analytic
5) Clean/Environmental control pod with real-time energy and environmental monitoring and analytic
6) Capacity and Throughput Simulation

Any Mix Any Volume-X (AMAv-X) Line : Milestones

Phase 1: Q3/25 ~ Q1/26 (MVP Showcase)






- Old Assets disposal
- Existing equipment refurbishment & relocation

- Zones 1 & 2: Deployment
- Zone 5: Partial Deployment (Automated Pill Dicer, RAISE)

Soft launch at BTM

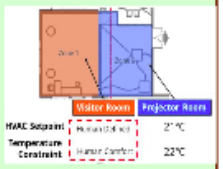
Full launch at BTM

Mar 25 | Sep 25 | Dec 25 | Mar 26 | Jun 26 | Sep 26

Phase 2: Q1/26 ~ Q3/26 (Full Showcase)

- Zone 3: Procure Continuous Tablet Dispensing
- Zone 4: Low-Code Connectivity and Dashboard
- Zone 5: Develop Automated QC Process








- Zone 3: Deploy Continuous Tablet Dispensing
- Zone 4: Deploy Digital Control Room
- Zone 5: Deploy Automated QC Process

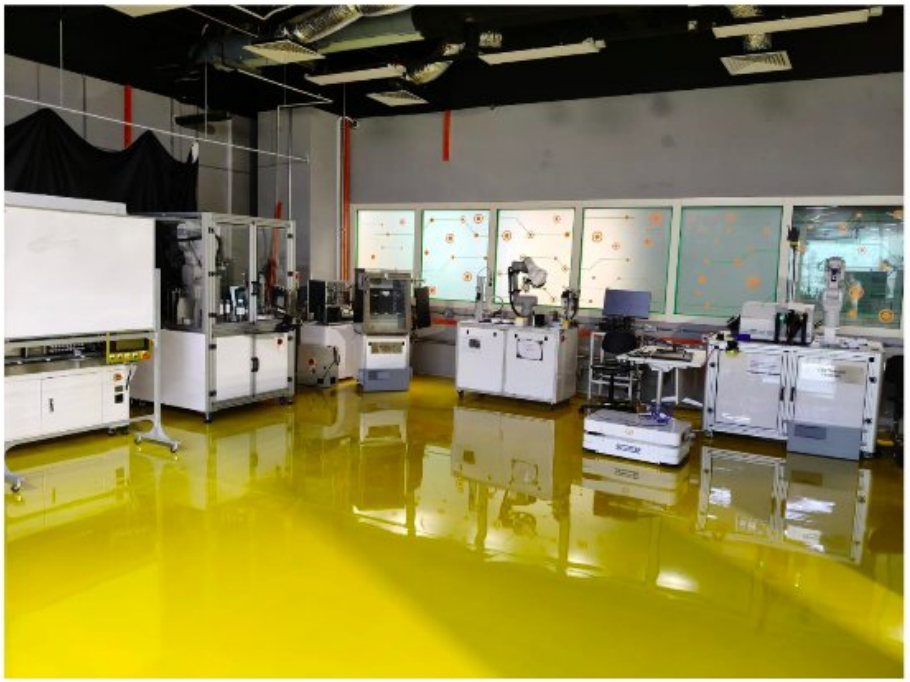
(Security / Sensitivity) Classification

Source/Reference:

Photos for AMAV-X Line



Zone 4: Digital Control Centre

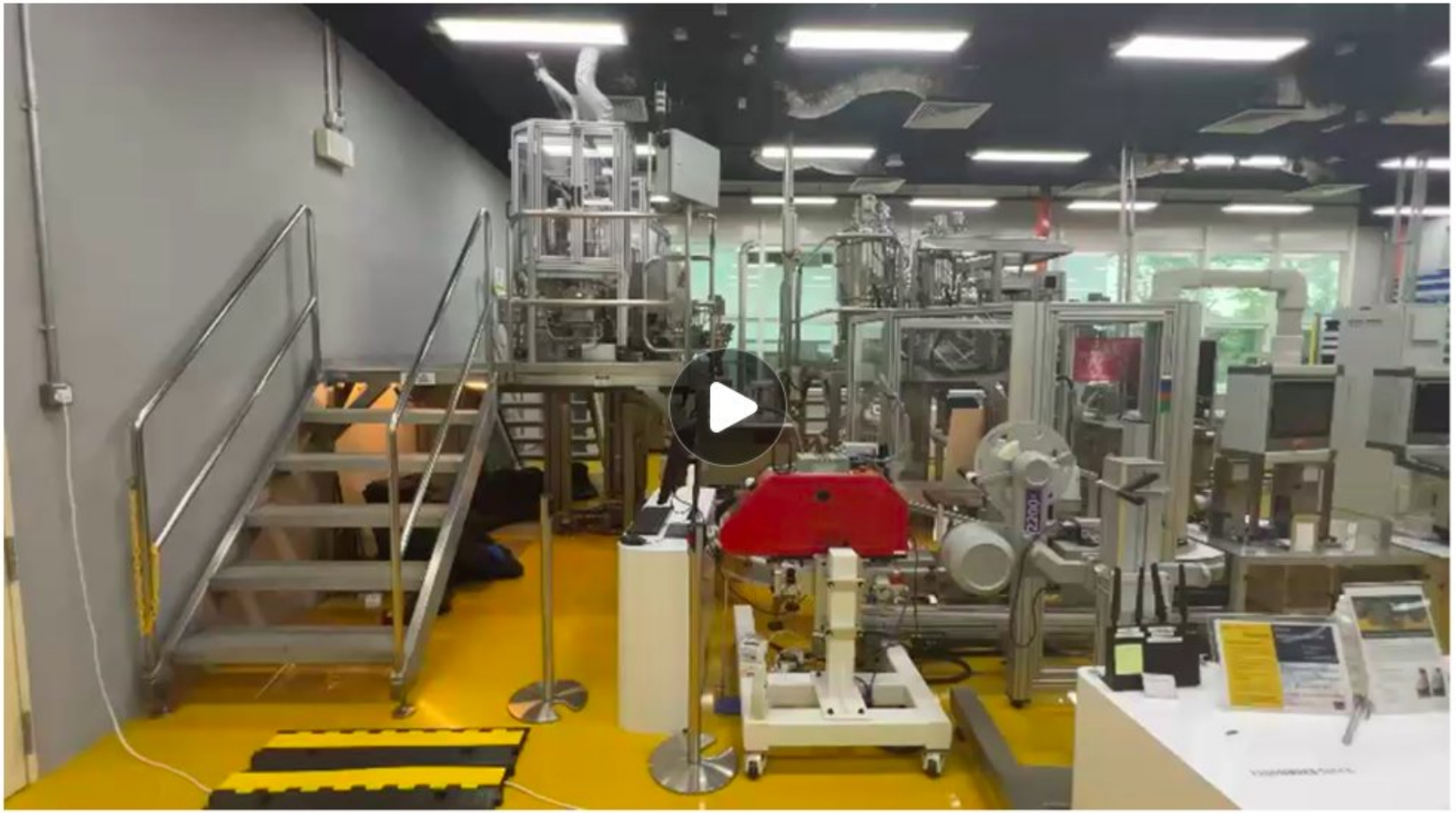


Zone 5: Lab of the Future

(Security / Sensitivity) Classification

Source/Reference:

Videos for AMAV-X Line



AMAV-X Line

(Security / Sensitivity) Classification

Source/Reference:

Technology Highlights

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(Security / Sensitivity) Classification

AI-Powered Prescriptive Maintenance with Limited Data



Alok Verma
Technical Lead



Manikanda
Technical Member

Collaborators
Motionwell AUTOMATION

Current Practice

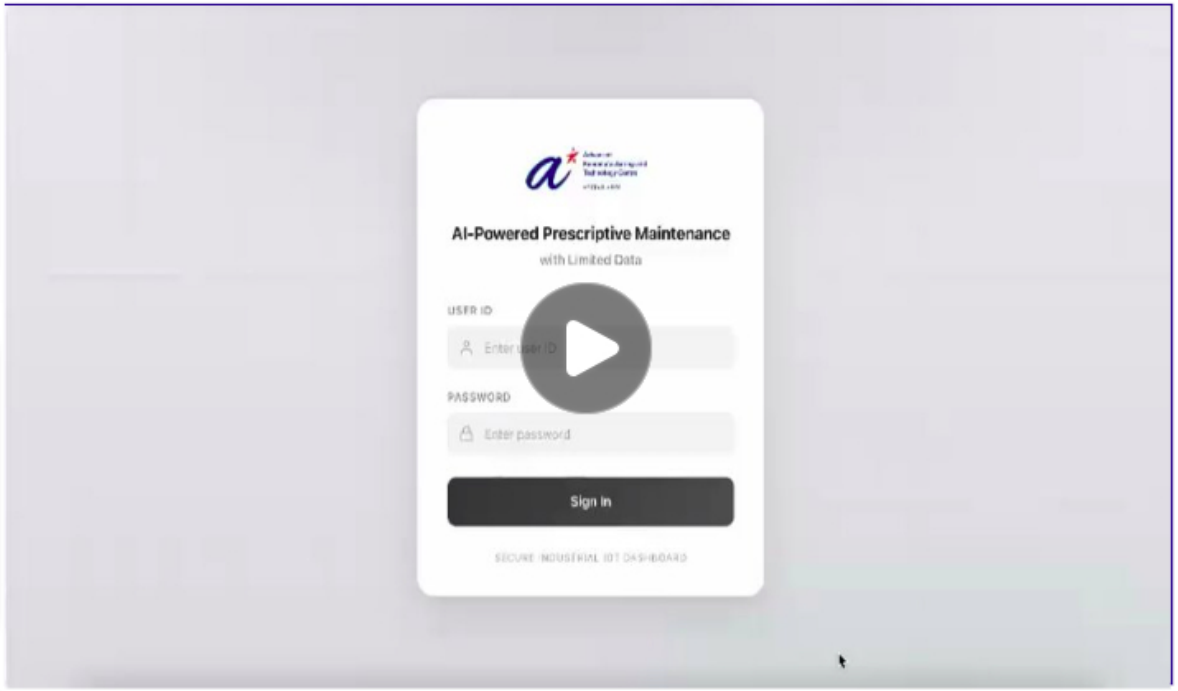
- Scheduled/ Reactive Maintenance
- Unplanned downtime and insufficient planning→ Higher Cost
- Any-Mix-Any-Volume , shorter production run → limited data
- Conventional Model Training needs high volume of production data to achieve high accuracy

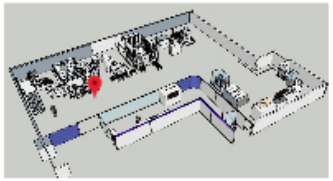
ARTC Solutions

- Develop AI model for precision predictive maintenance that deliver high accuracy **despite limited data**.
- **Real-Time Monitoring** with Digital Integration and Performance Visualization
- AI-powered early anomaly detection using **common model** and decision- making capability
- **Prescribe action** before machine downtime

Business Impact

- Improve accuracy on predictive maintenance and accelerate ML capability for prescriptive maintenance
- ~ **15% Increase** in Productivity
- ~ **20% Reduction** in Time for Troubleshooting
- ~ **15% Reduction** in Unplanned Machine Downtime





AI Common Model Predictive Quality Monitoring for Tablet Filling and Packaging Line

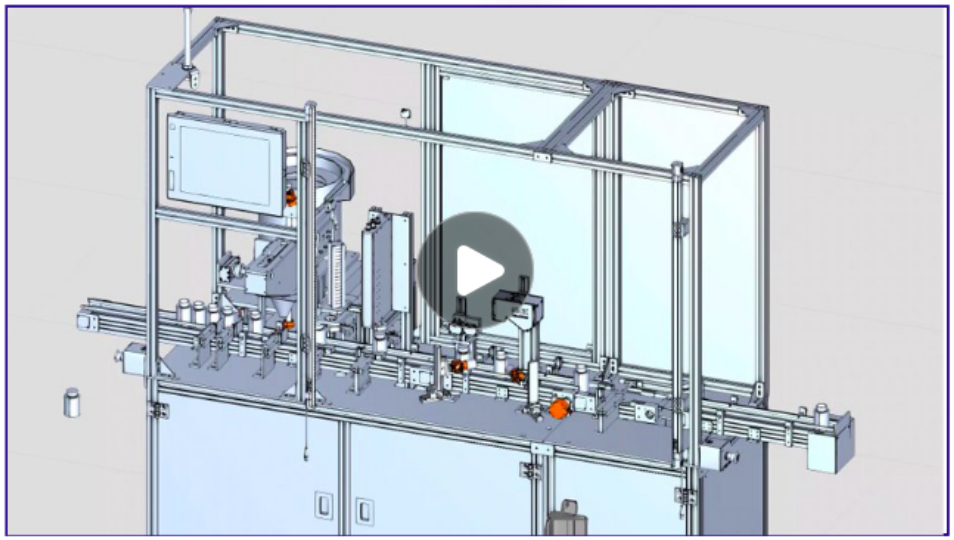



Alok Verma Manikanda
Technical Lead

Collaborators
SICK

Current Practice

- Manual sample inspections result in slower inspection speeds, inconsistent judgments, reduced detection accuracy, higher operational costs, and potential loss of customer trust

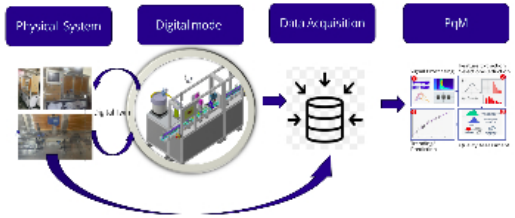


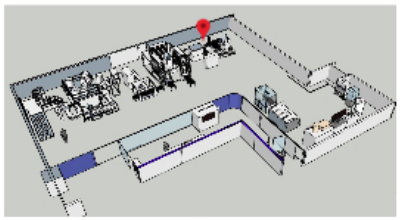
ARTC Solutions

- Develop **AI common model** and algorithm for predictive quality that deliver high accuracy **despite limited data**
- AI-powered quality monitoring for tablet inspection, dispensing, and capping quality process

Business Impact

- Enable real-time quality monitoring and prediction
- 100% online quality inspection
- Dashboard for quality reporting and production performance





Automated GMP Compliant Cleaning and Disinfection Technology



Technical Lead
Uma Sagar

Collaborators

BIONTECH GSK sanofi BIOPIPS

Current Practice

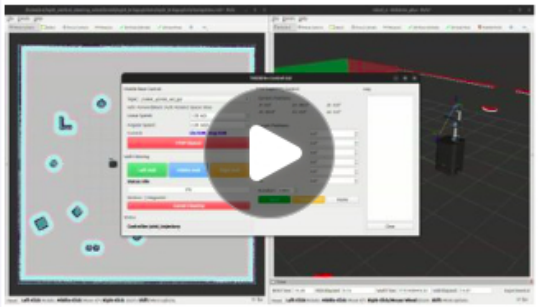
- Cleaning in cleanrooms is labor-intensive
- Exposes operators to cleaning chemicals
- Struggles to meet the consistency in maintaining Good Manufacturing Practices (GMP) standards.
- Data logging is manual

ARTC Solutions

- All robotic operations are managed through the Robot Management Framework 2 (RMF2)
- Robotic cleaning ensured comprehensive surface coverage while complying with Good Manufacturing Practice (GMP) standards
- A **dedicated logging system** captures key operational data such as cleaning duration, area coverage, and frequency

Business Impact

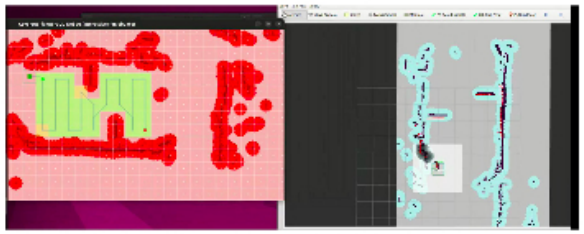
- **Repeatable cleaning** performance
- Improved operational scalability
- Data-driven process optimization
- **Full traceability** and faster audit reporting
- Sustainable cost reduction



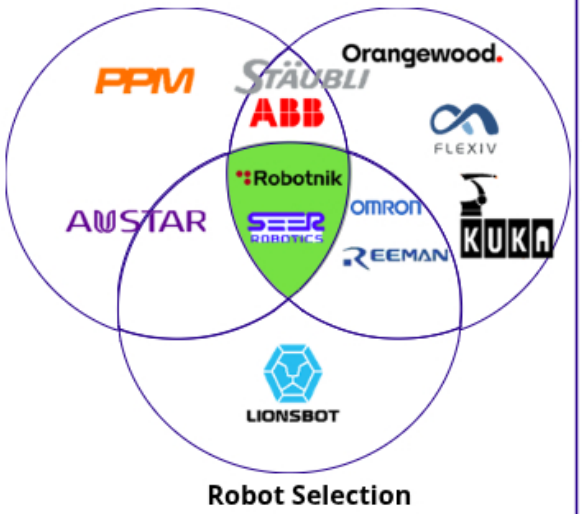
Vertical Cleaning



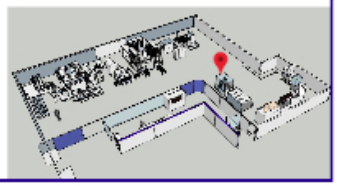
Floor Cleaning Robot



Path coverage for floor Cleaning



Robot Selection



AMAv-X Smart Liquid – Liquid Mixing Homogeneity Control



Technical Lead
Sean Ang

Collaborators

SING DYNAMIC

Problem statement

- Extensive **study and validation** to determine mixing time required for **each liquid formulation** type
- With fixed mixing time and post-mixing QC verification, there are challenges with **under and overmixing**

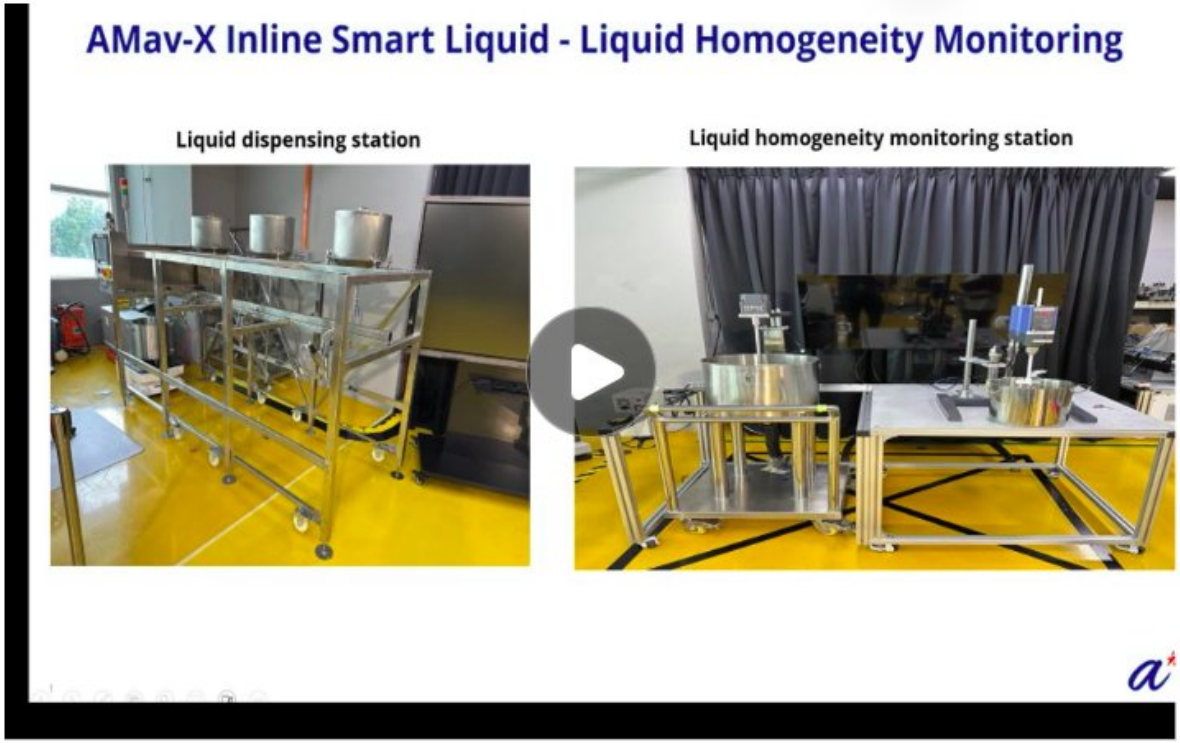
ARTC Solution

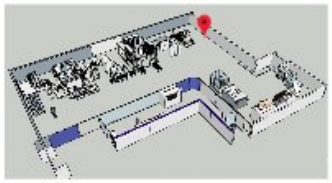
Smart Liquid Homogeneity system

- Multi-sensor (Electromagnetic / Reflectometry) data acquisition to capture liquid properties
- Smart Algorithm to enable live monitoring, aligned to QC testing requirements
- Real-time quality feedback on mixing status with visualizations

Business Benefits

- Increased productivity and capacity for mixing process through process optimization
- Timesaving of up to 80% for mixing process





Energy and Climate Management for Cleanroom Operations



Technical Lead
Vishal

Collaborators
Motionwell AUTOMATION

Current Practice

- Current cleanroom operations are managed independently such as energy, HVAC and operational data.
- Current cleanroom operations are controlled based on the objective to maintain air changes / hour which can result in overcooling and excess energy consumption.

Our Solution

- Real-time visibility of energy utilization and climate properties
- Using data-driven methods to simulate cleanroom operations with energy utilization of HVAC systems
- Provide recommendations to balance energy usage and climate requirements to adapt to manufacturing demand.

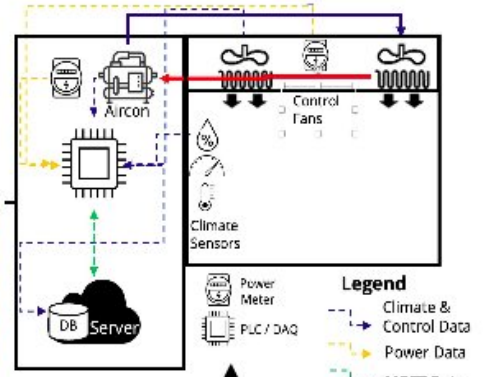
Business Benefits

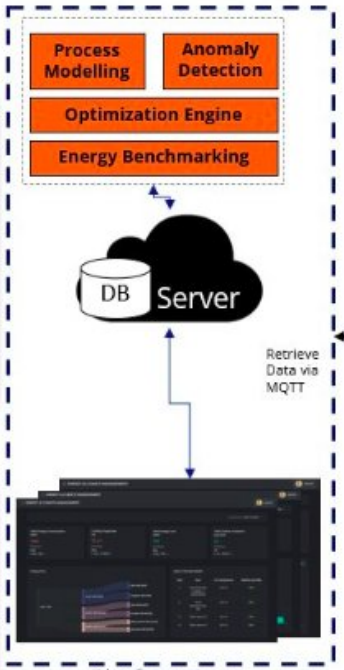
- A dashboard to enable **real-time monitoring** of key parameters for monitoring of climate, energy and operational parameters for cleanroom compliance.
- Identify **key correlations** between operations and energy utilization of facility systems to identify **optimization opportunities**.
- Provide **control recommendations** towards adaptive facility operations to match production operations.



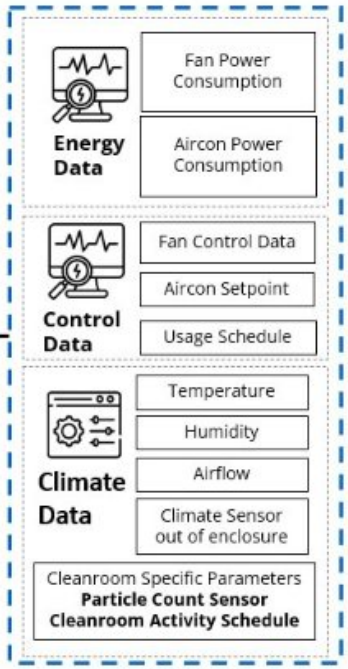








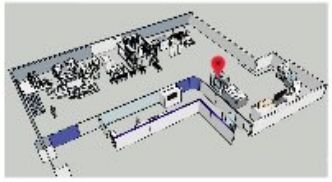
Platform Layer



Data Layer

ECMaS Platform and Data Layer





Liquid Pouch Work Order Management & Automated Fulfilment with Mendix



Technical Lead
Tan Peng Yu

Collaborators

SIEMENS

opex Consulting Group

Current Practice

- **Manually place work orders** using HMIs at each individual stations
- Material transfer via mobile robots are **triggered manually**
- No automated start-to-end production
- **No real-time** production and WO data collection/tracking

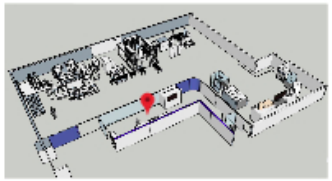
ARTC Solutions

- Clean and user-friendly foundation UI for work order management, with user access management
- **Dashboard for production tracking** of process steps statuses
- **RMF2.0 integration** into workflow to trigger for material transfer automatically
- **Production line triggered automatically** from UI when work order is placed

Business Benefits

- Accelerated order fulfillment with end-to-end automated work order orchestration and material flow
- Higher operational efficiency through **real-time production visibility** and reduced manual intervention
- Stronger decision-making control via a centralized Nerve Centre turning **shopfloor data into actionable insights**





Lab Tour for Physical Demos

Sub-title: Open Sans, regular, 24pt, left align



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